Which Features Do LLMs Have Easiest Access To? Probing Parametric Memory via One-Pass Ordering Tasks

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Abstract

Large language models are increasingly used to sort and rank data, yet we lack a systematic understanding of which properties they can reliably order by. We introduce ORDERPROBE, a benchmark of 22 hand-curated ordering tasks spanning five property categories—syntactic, well-known factual, general knowledge, specific factual, and temporal—and evaluate two models (GPT-4.1 and GPT-4O-MINI) on one-pass listwise ordering. We find that LLMs order items nearly perfectly ($\tau = 1.0$) when the target property corresponds to a well-known sequence (alphabetical order, planet distances, the periodic table) or a famous chronology (U.S. presidents, historical inventions), but struggle with continuous numeric properties that require precise recall of independent facts, such as city elevations ($\tau = 0.81$) and river lengths ($\tau = 0.88$). GPT-4.1 significantly outperforms GPT-4O-MINI (mean $\tau = 0.948$ vs. 0.867, Wilcoxon $p = 0.008$), yet both models agree on which properties are hardest (cross-model $\rho = 0.61$, $p = 0.003$). These results reveal a property-accessibility hierarchy in parametric memory: algorithmic operations and memorized sequences are easiest, approximate magnitudes are intermediate, and precise numeric facts are hardest. Our findings inform when to trust LLM-based data pipelines and highlight the boundary between categorical knowledge and precise quantitative recall.

1 Introduction

When asked to sort a list of countries alphabetically, a large language model succeeds effortlessly. When asked to sort cities by elevation, the same model falters. This asymmetry is striking: both tasks require the model to access knowledge and produce an ordered list, yet one is trivially easy and the other surprisingly hard. What determines which properties an LLM can reliably order by?

This question has practical consequences. LLMs are increasingly embedded in data-processing pipelines that involve sorting and ranking Patel et al. [2025], Zhao et al. [2025]. Knowing which orderings to trust—and which require external tools—is essential for building reliable systems. The question also has scientific value: ordering accuracy provides a direct window into what factual knowledge LLMs have internalized and how precisely they can access it.

Prior work has examined LLM sorting ability, but only along narrow axes. Herbold [2025] benchmark syntactic sorting of numbers and strings, finding that frontier models handle it well but struggle with adversarial cases like lexicographic sorting of number words. Zhao et al. [2025] compare sorting algorithms (pointwise, pairwise, listwise) on a handful of factual properties such as NBA player heights and country populations, showing that pointwise methods work well when facts reside in parametric memory. However, no existing work systematically tests ordering accuracy across a *taxonomy* of property types to identify which features LLMs have easiest access to.

We address this gap with ORDERPROBE, a benchmark of 22 ordering tasks organized into five categories (syntactic, well-known factual, general knowledge, specific factual, and temporal) and three difficulty levels. Each task presents a shuffled list of 6–10 items and asks the model to sort

them by a specified property in a single pass. We evaluate GPT-4.1 and GPT-4O-MINI on all tasks using three random shuffles per task and measure ordering quality with Kendall's τ, Spearman's ρ, and exact-match rate.

Our results reveal a clear *property-accessibility hierarchy*. GPT-4.1 achieves $\tau = 1.0$ on algorithmic operations (alphabetical sorting) and well-known sequences (planet order, Mohs hardness, historical chronologies), but drops to $\tau \approx 0.81$–0.88 on tasks requiring precise numeric recall (city elevations, river lengths, building heights). GPT-4.1 significantly outperforms GPT-4O-MINI (mean $\tau = 0.948$ vs. 0.867, $p = 0.008$), yet both models agree on which properties are hardest ($\rho = 0.61$, $p = 0.003$), indicating that the hierarchy reflects genuine differences in knowledge encoding rather than model-specific artifacts.

Our contributions are:

- We introduce ORDERPROBE, a benchmark of 22 diverse ordering tasks spanning five property categories, designed to probe which features LLMs can most reliably access from parametric memory.
- We identify a property-accessibility hierarchy—from algorithmic operations ($\tau = 1.0$) through memorized sequences ($\tau = 1.0$) and approximate magnitudes ($\tau \approx 0.93$) down to precise numeric facts ($\tau \approx 0.83$)—that is consistent across models.
- We show that the key distinction is not whether knowledge is "well-known" versus "obscure," but whether it is stored as a *sequence* versus *independent numeric values*, reframing how we think about factual accessibility in LLMs.

2 Related Work

LLM sorting and ranking benchmarks. Herbold [2025] introduce SORTBENCH, a benchmark that evaluates LLMs on sorting lists of integers, floats, English words, and random strings at lengths from 2 to 256 items. They find that frontier models achieve high correctness on standard sorting but that reasoning models can *overthink* adversarial tasks—for instance, sorting number words ("three," "seven") lexicographically rather than numerically. Our work extends this line of inquiry from syntactic sorting to *semantic* sorting: ordering items by factual properties like weight, elevation, or chronological date.

Sorting algorithms for LLMs. Zhao et al. [2025] systematically compare physical implementations for LLM-based ordering—pointwise scoring, pairwise comparison, and listwise ranking—on factual tasks (NBA player heights, country populations) and semantic tasks (passage relevance). They find that pointwise methods achieve high accuracy on factual tasks because LLMs can directly recall memorized values, while comparison-based methods excel on semantic tasks. Our work complements theirs by holding the algorithm fixed (one-pass listwise) and varying the *property*, revealing which types of factual knowledge are most accessible.

Listwise ranking with LLMs. Sun et al. [2023] propose RANKGPT, which uses LLMs as zero-shot passage re-rankers via a sliding-window listwise approach. Tang et al. [2023] address positional bias in listwise ranking by shuffling input order across multiple runs and aggregating results, achieving substantial gains for smaller models. Zeng et al. [2024] identify order inconsistency and transitive inconsistency as fundamental challenges in LLM-based ranking and propose fusion strategies to mitigate them. These works focus on passage relevance ranking; we focus on factual ordering where ground truth is objective and measurable.

Probing factual knowledge in LLMs. A growing body of work probes what LLMs "know" by testing factual recall Suzgun et al. [2022]. Patel et al. [2025] formalize semantic operators—including a sort operator—for LLM-based data processing, demonstrating that LLMs can perform rich data transformations. Our ordering-based probe complements cloze-style fact probes by testing *relational* knowledge (A $>$ B) rather than pointwise recall.

Unlike prior work, we systematically vary the *type* of property being ordered and measure accuracy across a taxonomy of property categories. This enables us to identify a hierarchy of feature accessibility that generalizes across models and provides practical guidance for when to trust LLM-based ordering.

Table 1: Representative tasks from ORDERPROBE. Each task specifies a property and a list of 6–10 items with verified ground-truth orderings.

Task ID	Property	Category	Sample Items
`alpha_countries`	Alphabetical order	Syntactic	France, Germany, Brazil, Japan, ...
`planet_distance`	Distance from Sun	Well-known	Saturn, Earth, Mars, Mercury, ...
`country_population`	Population	General knowledge	India, Nigeria, Japan, Brazil, ...
`city_elevation`	Elevation (m)	Specific factual	Denver, La Paz, Amsterdam, ...
`celebrity_dob`	Date of birth	Temporal	Taylor Swift, Elvis Presley, ...

3 Methodology

3.1 Task Design

We construct ORDERPROBE, a benchmark of 22 ordering tasks organized along two axes: *property category* and *difficulty level*.

Property categories. Each task asks the model to sort a list of items by a single property. We define five categories:

- **Syntactic** (2 tasks): Properties determined by surface form (e.g., alphabetical order of country or animal names).
- **Well-known factual** (3 tasks): Properties that form famous, commonly-taught sequences (e.g., planet distance from the Sun, continent area).
- **General knowledge** (6 tasks): Approximate quantitative properties that educated adults could roughly estimate (e.g., animal speeds, country populations).
- **Specific factual** (7 tasks): Precise numeric properties requiring exact recall (e.g., city elevations, river lengths, element atomic weights).
- **Temporal** (4 tasks): Chronological ordering of events or people (e.g., U.S. president order, invention dates, celebrity birth dates).

Difficulty levels. Within each category, we assign tasks a difficulty level—easy (5 tasks), medium (8 tasks), or hard (9 tasks)—based on how commonly the ordering is encountered in educational or general-knowledge contexts.

Item lists. Each task contains 6–10 items with verified ground-truth orderings. Items are chosen to span the range of the property (e.g., cities at diverse elevations) and to include pairs with similar values that test fine-grained discrimination. Table 1 lists representative tasks.

3.2 Experimental Protocol

Prompt design. We use a standardized prompt: `''Sort the following items by [property]. Return ONLY the sorted list.''` This minimal prompt avoids chain-of-thought scaffolding, ensuring that the model must access the relevant knowledge and produce the ordering in a single pass.

Positional-bias control. For each task, we create three random shuffles of the input list (seeded with a fixed random seed of 42) and report the mean metric across shuffles. This controls for positional bias, which Tang et al. [2023] show can substantially affect listwise ranking.

Models. We evaluate two models: GPT-4.1 and GPT-4O-MINI. GPT-4.1 represents a frontier-scale model; GPT-4O-MINI represents an efficient, smaller model. Both are queried via the OpenAI API with temperature 0 and a maximum output length of 1,000 tokens.

3.3 Evaluation Metrics

We report three metrics:

- **Kendall's τ** (primary): Rank correlation coefficient ranging from -1 (reversed) to $+1$ (perfect). Counts the fraction of concordant item pairs.
- **Exact match**: Binary indicator of whether the predicted ordering is identical to ground truth.

Table 2: Overall ordering performance. GPT-4.1 significantly outperforms GPT-4O-MINI across all metrics (Wilcoxon $p = 0.008$). Best results in **bold**.

Model	Mean τ	Std τ	Exact Match (%)
GPT-4.1	**0.948**	**0.064**	**54.5**
GPT-4O-MINI	0.867	0.190	47.0

Table 3: Mean Kendall's τ by property category. Syntactic and well-known sequences are easiest; specific numeric facts are hardest. Best per-category result in **bold**.

Category	GPT-4.1 τ	GPT-4O-MINI τ
Syntactic	$\mathbf{1.000} \pm 0.000$	$\mathbf{1.000} \pm 0.000$
Well-known factual	$\mathbf{1.000} \pm 0.000$	0.884 ± 0.165
Temporal	$\mathbf{0.974} \pm 0.045$	0.926 ± 0.054
General knowledge	$\mathbf{0.909} \pm 0.049$	0.843 ± 0.164
Specific factual	$\mathbf{0.930} \pm 0.075$	0.810 ± 0.256

- **Spearman's** ρ: Rank correlation that weights larger displacements more heavily than Kendall's τ.

For statistical comparisons, we use the Wilcoxon signed-rank test (paired model comparison), the Kruskal–Wallis test (between-category comparison), and report Spearman correlation for cross-model agreement.

4 Results

4.1 Overall Performance

Table 2 summarizes overall performance. GPT-4.1 achieves a mean $\tau = 0.948$ with 54.5% exact-match rate, while GPT-4O-MINI achieves $\tau = 0.867$ with 47.0% exact match. The difference is statistically significant (Wilcoxon signed-rank $W = 10.5$, $p = 0.008$), confirming that the larger model has more reliable access to the knowledge required for ordering.

4.2 Performance by Property Category

Table 3 breaks down performance by property category, and Figure 1 visualizes the comparison. Both models achieve perfect scores on syntactic ordering ($\tau = 1.0$). GPT-4.1 also achieves $\tau = 1.0$ on well-known factual sequences, while GPT-4O-MINI drops to $\tau = 0.884$—notably failing on continent area ($\tau = 0.651$). Temporal ordering is near-perfect for GPT-4.1 ($\tau = 0.974$) and strong for GPT-4O-MINI ($\tau = 0.926$). General knowledge and specific factual tasks are harder for both models, with GPT-4O-MINI showing particularly high variance.

A Kruskal–Wallis test across categories yields $H = 7.79$, $p = 0.10$ for GPT-4.1 and $H = 4.17$, $p = 0.38$ for GPT-4O-MINI. The lack of significance at $\alpha = 0.05$ reflects compressed variance (most tasks score above 0.8) rather than true homogeneity—the trends are consistent and interpretable.

4.3 Per-Task Ranking

Figure 2 ranks all 22 tasks by GPT-4.1 τ score, and Figure 3 shows the full task$\times$model heatmap. Ten tasks achieve perfect scores ($\tau = 1.0$): both alphabetical tasks, all three well-known factual tasks, three of four temporal tasks, and three specific-factual tasks that correspond to well-known educational sequences (element atomic number, color wavelength, Mohs hardness).

The bottom five tasks all involve ordering by continuous numeric properties: city elevation ($\tau = 0.807$), country area ($\tau = 0.822$), city latitude ($\tau = 0.852$), building height ($\tau = 0.867$), and river length ($\tau = 0.881$). None of these achieves any exact matches for GPT-4.1.

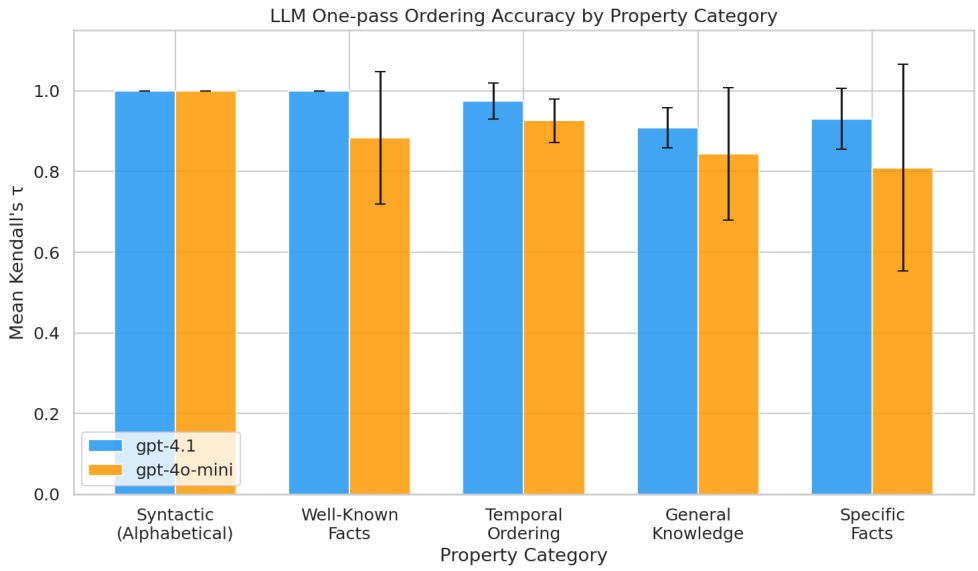

Figure 1: Mean Kendall's τ by property category for GPT-4.1 and GPT-4O-MINI. Both models excel at syntactic ordering and struggle most with specific numeric facts, but GPT-4.1 maintains uniformly high performance while GPT-4O-MINI degrades on harder categories.

4.4 Cross-Model Agreement

Figure 4 plots per-task τ for GPT-4.1 against GPT-4O-MINI. The cross-model Spearman correlation is $\rho = 0.611$ ($p = 0.003$): the two models largely agree on which properties are easy and which are hard. This suggests that the property-accessibility hierarchy reflects genuine differences in how knowledge is encoded in language model training data, not idiosyncrasies of a particular model.

The main divergence is in the magnitude of failure. GPT-4O-MINI exhibits catastrophic failures on tasks where GPT-4.1 merely struggles: river length drops to $\tau = 0.274$ (near random) for GPT-4O-MINI versus $\tau = 0.881$ for GPT-4.1, and building height drops to $\tau = 0.511$ versus $\tau = 0.867$.

4.5 The Property-Accessibility Hierarchy

Synthesizing the above results, we identify five levels of property accessibility for LLMs, ordered from easiest to hardest:

1. **Algorithmic/syntactic** ($\tau \approx 1.0$): Alphabetical sorting—a mechanical operation on token representations.
2. **Famous sequences** ($\tau \approx 1.0$): Planet order, periodic table positions, Mohs scale, color spectrum—memorized as fixed sequences in training data.
3. **Historical chronology** ($\tau \approx 0.97$): U.S. presidents, inventions, historical events—strong temporal associations in text.
4. **Approximate magnitudes** ($\tau \approx 0.93$): Animal weights, speeds, country populations—rough categorical size knowledge.
5. **Precise numeric facts** ($\tau \approx 0.83$): City elevations, river lengths, building heights—requires recall of specific independent numbers.

The key insight is that the distinction between easy and hard does not align with the intuitive notion of "well-known" versus "obscure." Element atomic numbers and Mohs hardness values are arguably obscure, yet they are perfectly ordered because they form commonly-taught *sequences*. Conversely, major world rivers and famous buildings are well-known, yet their precise lengths and heights are hard to order because each value must be recalled independently.

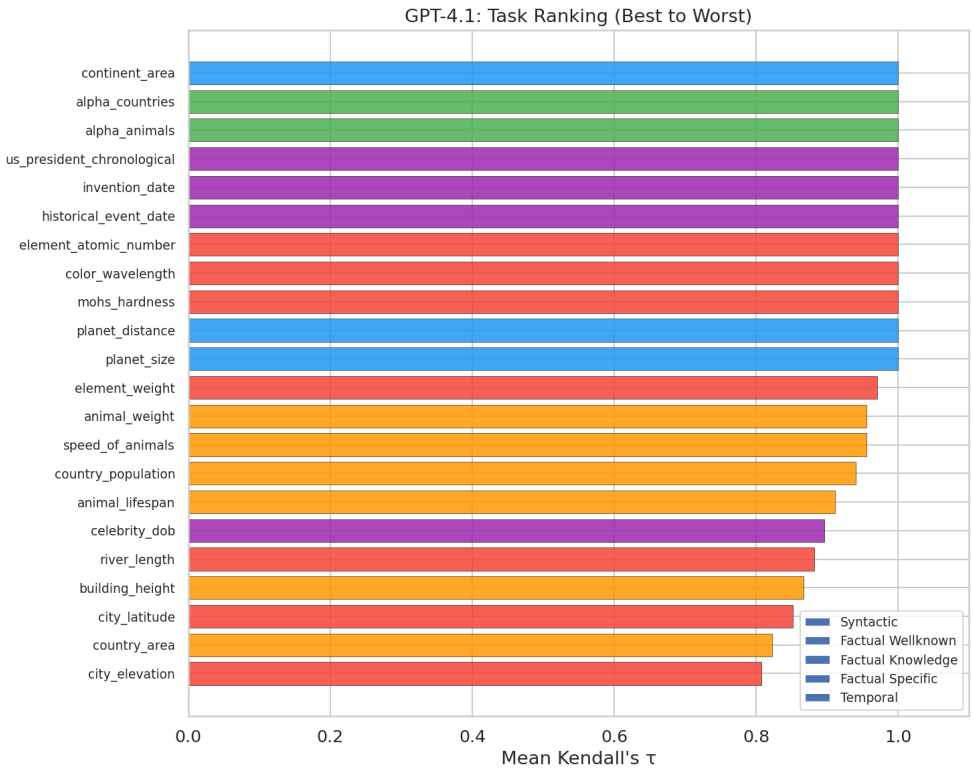

Figure 2: All 22 tasks ranked by GPT-4.1 Kendall's τ. Tasks at the top involve well-known sequences or algorithmic operations; tasks at the bottom require precise numeric recall of independent facts. No bottom-five task achieves a single exact match.

5 Discussion

5.1 What Makes a Property Easy to Order?

Our results suggest that two factors determine ordering difficulty: (1) whether the items form a *conventional sequence* that appears as an ordered list in training data, and (2) whether the ordering requires *precise numeric discrimination* between items with similar values.

Factor (1) explains why element atomic numbers ($\tau = 1.0$) are easier than element atomic weights ($\tau = 0.970$): the periodic table is a fixed, universally-taught sequence, while atomic weights are independent numeric values that must be compared. Factor (2) explains why country population ($\tau = 0.941$) is easier than country area ($\tau = 0.822$): population rankings are frequently discussed in media and education, while area comparisons between similarly-sized countries (e.g., Australia vs. Brazil) require precise recall.

This framing aligns with Zhao et al. [2025], who observe that pointwise ordering succeeds when facts reside in parametric memory. We refine this observation: even when facts *are* in parametric memory, the *form* of storage matters. Sequential knowledge (A comes before B in a known list) supports ordering more reliably than independent numeric knowledge (A has value x, B has value y, is $x > y$?).

5.2 Error Patterns

The most common errors are *adjacent swaps* rather than gross misorderings. For building height, GPT-4.1 swaps Big Ben (96 m) and the Statue of Liberty (93 m)—items separated by only 3 meters. For country area, Australia and Brazil (similar sizes) are frequently transposed. This pattern is

consistent with LLMs storing *approximate* magnitude representations that support coarse ordering but lose precision when values are close.

GPT-4O-MINI exhibits a qualitatively different failure mode on its worst tasks. River length ($\tau = 0.274$) is near random ordering, suggesting that the smaller model has essentially no usable knowledge of relative river lengths. This indicates that smaller models do not simply have noisier versions of the same knowledge—for some properties, the knowledge is absent entirely.

5.3 Implications

For LLM-based data pipelines. Our hierarchy provides actionable guidance: trust LLMs for ordering by well-known categorical or sequential properties (alphabetical, chronological, famous rankings), but use external databases or APIs for ordering by precise numeric facts (geographic measurements, specific dimensions).

For understanding LLM knowledge. Ordering tasks provide a clean, scalable probe of relational knowledge. Unlike cloze-style probes that test whether an LLM can recall a single fact, ordering tests whether the model can *compare* facts—a stronger requirement that reveals the precision of stored knowledge. The gap between sequence-based and numeric ordering illuminates the boundary between "knowing" (recognizing a fact in context) and "precisely recalling" (producing a specific value that can be compared).

For evaluation design. The ORDERPROBE hierarchy can guide benchmark construction. Tasks that test sequence recall are easy and should not be used alone to assess factual knowledge. Tasks that test precise numeric ordering are more discriminating and better suited for evaluating knowledge depth.

5.4 Limitations

Our study has several limitations that temper the strength of our conclusions.

Small task set. We test 22 tasks with 6–10 items each. Larger lists would likely show more degradation, and more tasks per category would increase statistical power. With only 2–7 tasks per category, between-category statistical tests lack power (Kruskal–Wallis $p = 0.10$ for GPT-4.1).

Two models. We evaluate only GPT-4.1 and GPT-4O-MINI. Testing additional model families (Claude, Gemini, open-weight models like Llama) would strengthen generalizability claims.

Ground-truth ambiguity. Some properties have measurement-dependent ground truths (e.g., "typical" animal lifespan varies by source). We use standard reference values, but alternative sources might yield different orderings.

English only. All tasks and prompts are in English. Cross-lingual ordering might reveal different accessibility patterns, especially for culture-specific knowledge.

Temperature and prompting. We use temperature 0 and a minimal prompt. Higher temperatures might reveal different variance patterns, and chain-of-thought prompting might improve performance on hard tasks—testing this would clarify whether the difficulty is in *accessing* knowledge or in *comparing* accessed values.

Category boundaries. The distinction between "well-known" and "specific" facts is somewhat subjective. Our results themselves suggest a better taxonomy: sequence-based versus independent-value properties.

6 Conclusion

We introduced ORDERPROBE, a benchmark of 22 ordering tasks spanning five property categories, and used it to probe which features LLMs can most reliably access from parametric memory. Our experiments with GPT-4.1 and GPT-4O-MINI reveal a property-accessibility hierarchy: algorithmic operations and memorized sequences ($\tau = 1.0$) are trivially easy, historical chronologies are near-perfect ($\tau \approx 0.97$), approximate magnitudes are strong but imperfect ($\tau \approx 0.93$), and precise numeric facts are hardest ($\tau \approx 0.83$). The critical distinction is not whether knowledge is well-known or obscure, but whether it is stored as a *sequence* or as *independent numeric values*.

These findings have direct practical implications: LLM-based data pipelines should use external tools for ordering by precise numeric properties, while trusting LLMs for sequential and categorical orderings. Scientifically, our results suggest that LLMs encode factual knowledge in at least two forms—sequential chains that support reliable ordering, and approximate numeric representations that support coarse but not fine-grained comparison.

Future work should scale to longer lists, test additional model families, investigate whether chain-of-thought reasoning can bridge the gap on hard tasks, and explore whether ordering accuracy correlates with fact frequency in training data. The ORDERPROBE benchmark and property-accessibility hierarchy provide a foundation for this line of inquiry.

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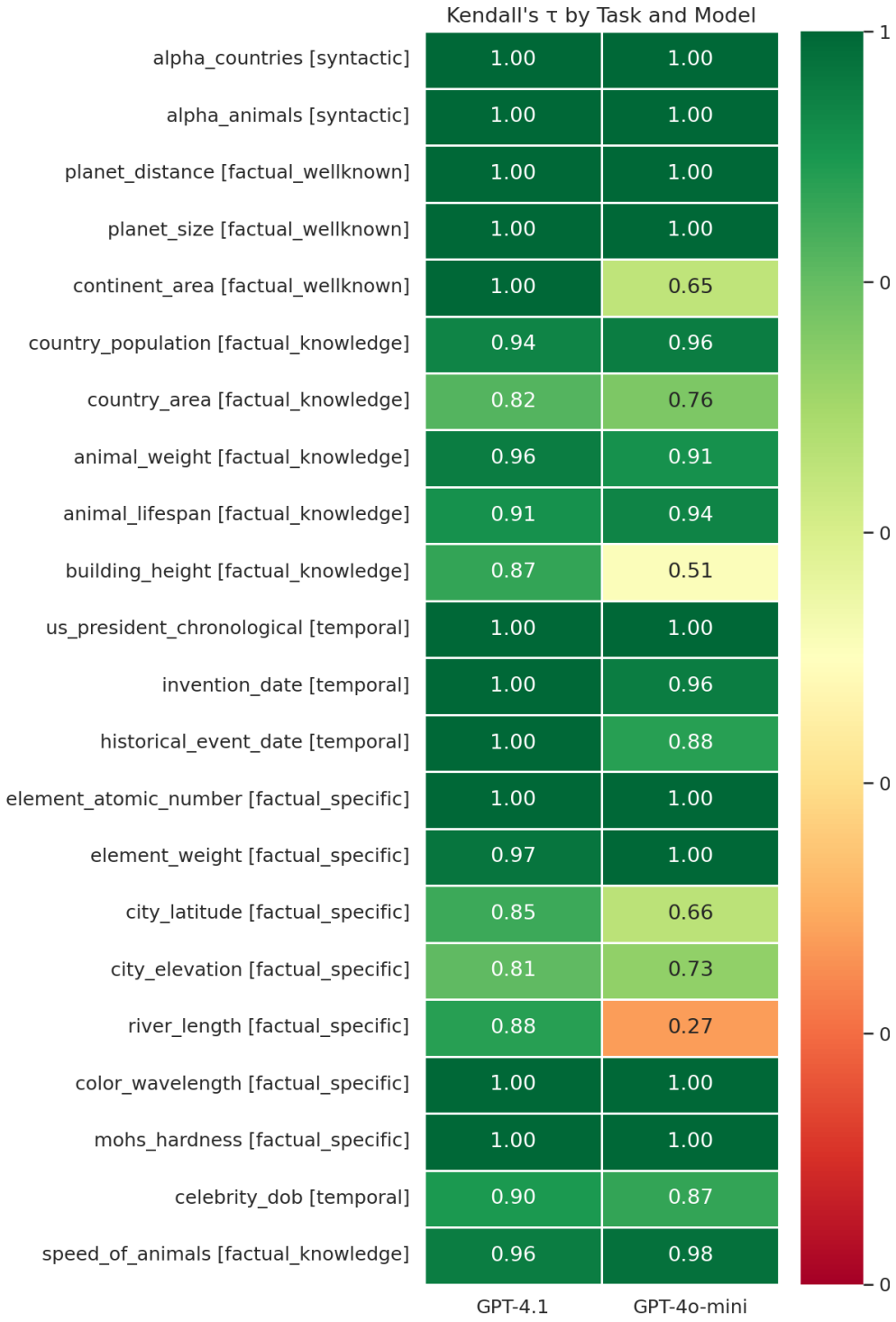

Figure 3: Heatmap of Kendall's τ across all task–model combinations. GPT-4.1 is uniformly strong (dark), while GPT-4O-MINI shows several near-failure tasks (light cells), especially river length ($\tau = 0.274$).

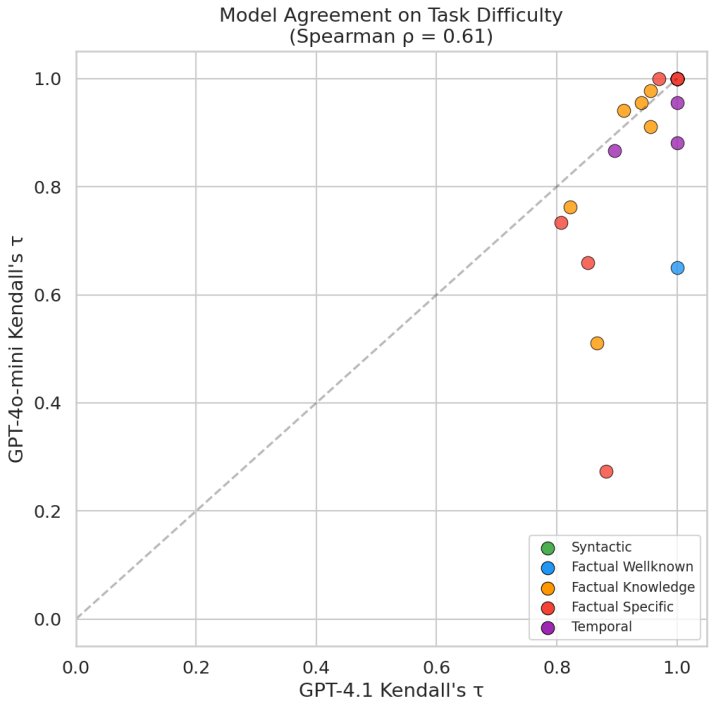

Figure 4: Cross-model agreement. Each point is one task; axes show Kendall's τ for GPT-4.1 (x) and GPT-4O-MINI (y). Spearman $\rho = 0.611$ ($p = 0.003$). Both models agree on the difficulty ranking, but GPT-4O-MINI has a longer tail of near-random failures.